Part Two

THE BEGINNING OF THE REVOLUTION

Twentieth-century physics has radically modified the Newtonian image of the world. The new steps are the basis of a great deal of today's technology. The deepening of our understanding of the world is based on two theories: general relativity and quantum mechanics. Both demand a daring re-evaluation of our conventional ideas about the world: space and time in relativity; matter and energy in quantum theory.

In this part of the book I describe the two theories in some detail, trying to clarify their core meaning and highlight the conceptual revolution they brought about. It's here that the magic of twentieth-century physics begins. Studying and trying to understand them in depth is a bewitching adventure.

These two theories – relativity and the quanta – provide the basis on which we are today building a quantum theory of gravity. They are the ground from which we are trying to move forwards.